A harmonic counterexample for the power-profile f-greedy question

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Source question

The source paper is Pablo M. Berna and Hung Viet Chu, *On Some Characterizations of Greedy-type Bases*, arXiv:2201.12029. For $f : \mathbb{N} \to \mathbb{R}$ and a finite set $A = \{a_1 < \cdots < a_m\}$, define

$$1_{f,A} = \sum_{r=1}^{m} f(r) e_{a_r}, \qquad \mathcal{D}_m^f(x) = \inf_{\substack{|A|=m \\ \alpha \in \mathbb{R}}} \|x - \alpha 1_{f,A}\|.$$

A basis is called f-greedy if there is C_f such that

$$\|x - \mathcal{G}_m(x)\| \le C_f \mathcal{D}_m^f(x) \qquad (x \in X,\ m \in \mathbb{N}). \tag{1}$$

The first future-research question in the source asks for a characterization of the functions f for which (1) is equivalent to ordinary greediness. The source proves the equivalence for regular f, constructs non-greedy examples for sparse f, and explicitly points to

$$f(n) = n^{-p}, \qquad p > 0,$$

as a class outside both results.

Counterexample theorem

Theorem 1. *For every $p > 0$, there is a normalized Schauder basis which is f_p-greedy for $f_p(n) = n^{-p}$ but is not greedy. In fact the same basis works for every $p > 0$.*

Idea of the proof

Use the harmonic norm

$$\|x\|_h = \max\left\{ \|x\|_\infty, \sum_{n=1}^{\infty} \frac{|x_n|}{n} \right\}.$$

Initial intervals have norm $1 + 1/2 + \cdots + 1/m$, while m coordinates far out have norm 1, so the unit vector basis is not democratic. The reason it is still n^{-p}-greedy is that the weighted tail

$$\sum_{r \ge q} \frac{r^{-p}}{r}$$

is comparable to q^{-p}. Thus the first missed f_p-coefficient pays for the whole remaining harmonic tail.

Proof

Let

$$X = \left\{ x = (x_n) \in c_0 : \sum_{n=1}^{\infty} \frac{|x_n|}{n} < \infty \right\}$$

with norm

$$\|x\| = \max \left\{ \sup_n |x_n|, \sum_{n=1}^{\infty} \frac{|x_n|}{n} \right\}. \tag{2}$$

The canonical unit vectors (e_n) form a normalized, 1-unconditional Schauder basis. Indeed, the coordinate projections are contractive, and partial sums converge in both the c_0 and weighted ℓ_1 parts of (2).

The basis is not democratic. If $A_m = \{1, \dots, m\}$, then

$$\|1_{A_m}\| = \sum_{n=1}^{m} \frac{1}{n}.$$

On the other hand, for each m one may choose N so large that $\sum_{n=N+1}^{N+m} 1/n \le 1$. Then

$$\left\| \sum_{n=N+1}^{N+m} e_n \right\| = 1.$$

The democracy constants therefore tend to infinity. Since the basis is unconditional, the Konyagin–Temlyakov characterization implies that it is not greedy.

It remains to prove f_p-greediness. Fix $p > 0$ and write

$$f(r) = r^{-p}.$$

Let $x \in X$, let Λ be the natural order-m greedy set of x, and put $y = x - P_\Lambda x$. We shall show that for every m-point set $A = \{a_1 < \cdots < a_m\}$ and every scalar α,

$$\|y\| \le \left(3 + \frac{2}{p}\right) \|x - \alpha 1_{f,A}\|. \tag{3}$$

Taking the infimum over A and α proves (1).

Set

$$z = x - \alpha 1_{f,A}, \qquad \delta = \|z\|.$$

First, $\|y\|_\infty \le \delta$. Indeed, let $i \notin \Lambda$. If $i \notin A$, then $y_i = x_i = z_i$. If $i \in A \setminus \Lambda$, then because $|A| = |\Lambda|$, there is $j \in \Lambda \setminus A$. Since Λ is greedy, $|x_j| \ge |x_i|$, and $z_j = x_j$; hence $|y_i| = |x_i| \le \delta$.

For the harmonic part, write

$$R = \{r \in \{1, \dots, m\} : a_r \notin \Lambda\}.$$

Then

$$\begin{aligned} \sum_{i \notin \Lambda} \frac{|x_i|}{i} &\le \sum_{i \notin \Lambda} \frac{|z_i|}{i} + |\alpha| \sum_{r \in R} \frac{r^{-p}}{a_r} \\ &\le \delta + |\alpha| \sum_{r \in R} \frac{r^{-p}}{a_r}. \end{aligned} \tag{4}$$

If $R = \varnothing$, (4) already gives the desired estimate. Otherwise let $q = \min R$. Since $a_q \in A \setminus \Lambda$, choose $j \in \Lambda \setminus A$. Then $|x_j| \geq |x_{a_q}|$, while

$$z_{a_q} = x_{a_q} - \alpha q^{-p}, \qquad z_j = x_j.$$

Consequently

$$\delta \geq \max\{|x_{a_q}|,\ |x_{a_q} - \alpha q^{-p}|\} \geq \frac{|\alpha| q^{-p}}{2}. \tag{5}$$

Also $a_r \geq r$ for every r, so

$$\sum_{r \in R} \frac{r^{-p}}{a_r} \leq \sum_{r=q}^{\infty} \frac{1}{r^{p+1}} \leq \left(1 + \frac{1}{p}\right) q^{-p}. \tag{6}$$

Combining (5) and (6) gives

$$|\alpha| \sum_{r \in R} \frac{r^{-p}}{a_r} \leq 2 \left(1 + \frac{1}{p}\right) \delta.$$

Together with (4),

$$\sum_{i \notin \Lambda} \frac{|x_i|}{i} \leq \left(3 + \frac{2}{p}\right) \delta.$$

The supremum estimate and the weighted estimate prove (3), so the basis is f_p-greedy. This completes the counterexample.

Verification notes

The proof is direct and does not use computation. The only source-paper input needed for context is the definition of f-greediness and the Konyagin–Temlyakov characterization of greedy bases as unconditional plus democratic.

The earlier partial packet

`solutions/partial/2201.12029_power_profile_log_democracy/`

is consistent with this counterexample: the present basis has precisely the harmonic logarithmic democracy growth that the partial theorem allowed.

Novelty check

The run indexes were searched for arXiv:2201.12029, f-greedy, $\mathcal{D}_m^f$, power-profile terminology, $1/n^p$, and the earlier partial packet. Web searches on 2026-06-21 for exact phrases around "f-greedy n^{-p} counterexample", "D_m^f harmonic greedy basis", and the source title with "$1/n^p$" found no later matching answer.

References

[1] P. M. Berna and H. V. Chu, *On Some Characterizations of Greedy-type Bases*, arXiv:2201.12029.

[2] S. V. Konyagin and V. N. Temlyakov, A remark on greedy approximation in Banach spaces, East J. Approx. 5 (1999), 365–379.